

A Correspondence with Freeman Dyson on Physics and Education

In March 2017, I wrote the essay “How to Teach Me Physics: Tradition Is Not Always a Virtue” (*Physics Today* **70**(3), 10–11 (2017)) for *Physics Today*. The piece was commissioned by the magazine’s editor and grew out of my own experiences—and difficulties—with the academic system in which I found myself at the time. After its publication, I sent the essay to Freeman Dyson by email. Below is his response. His words were deeply meaningful to me then, and they remain so today.

March 1, 2017

Dear Ricardo Heras,

Thankyou for the message and the Physics Today letter. I enjoyed reading the letter. I agree with you that the time spent in formal class-room lectures and course-work is mostly wasted. You don’t need all that stuff to do science. I always like to say that the best work I ever did in science was when I was most ignorant. My advice to you is to skip the classes as much as the system allows, and get to work on a real problem. When you work on a real problem, you quickly find out what you really need to know, and you find out whether you have what it takes to solve it.

I was tremendously lucky in several ways. First, I learned physics mostly from a wonderful book, “Theoretical Physics” by Georg Joos, which I still have on my shelf, still useful and amazingly up to date after eighty years. Second, because of wartime disruptions I escaped doing any physics courses as an undergraduate. Third, I arrived as a graduate student at Cornell and immediately met Bethe, my official supervisor, and Feynman who quickly became a friend. Fourth, I jumped into real problems with Bethe and Feynman and found useful things to do.

It is sad that only a few people can be as lucky as I was. But there are still plenty of opportunities for ignorant people to make important contributions. You just have to be ready to grab the opportunities when they arise. Do not imagine that you have to learn everything before you can do anything.

I will be glad to stay in touch. Wishing you plenty of luck, yours sincerely,

Freeman Dyson.